psymetadata: Social Identity Theory example

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2026-03-27

```r
library(metaselection)
library(metadat)
library(psymetadata)
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at C:/GitHub/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

Social Identity Theory Meta-analysis

```
dat_ex <- steffens2020

dat_ex <- dat_ex |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )

length(unique(dat_ex$studyid)); length(unique(dat_ex$esid))
```

```
## [1] 128
```

```
## [1] 251
```

Mean Only

Unadjusted estimate: mean only

```
ex_unadj_mean <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML"
)
ex_unadj_mean
```

```
##
## Random-Effects Model (k = 251; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0858 (SE = 0.0083)
## tau (square root of estimated tau^2 value):      0.2929
## I^2 (total heterogeneity / total variability):   95.96%
## H^2 (total variability / sampling variability):  24.77
##
## Test for Heterogeneity:
## Q(df = 250) = 6351.3935, p-val < .0001
##
## Model Results:
##
## estimate      se     zval    pval   ci.lb   ci.ub
##   0.4151  0.0193  21.5529  <.0001  0.3773  0.4528  ***
##
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

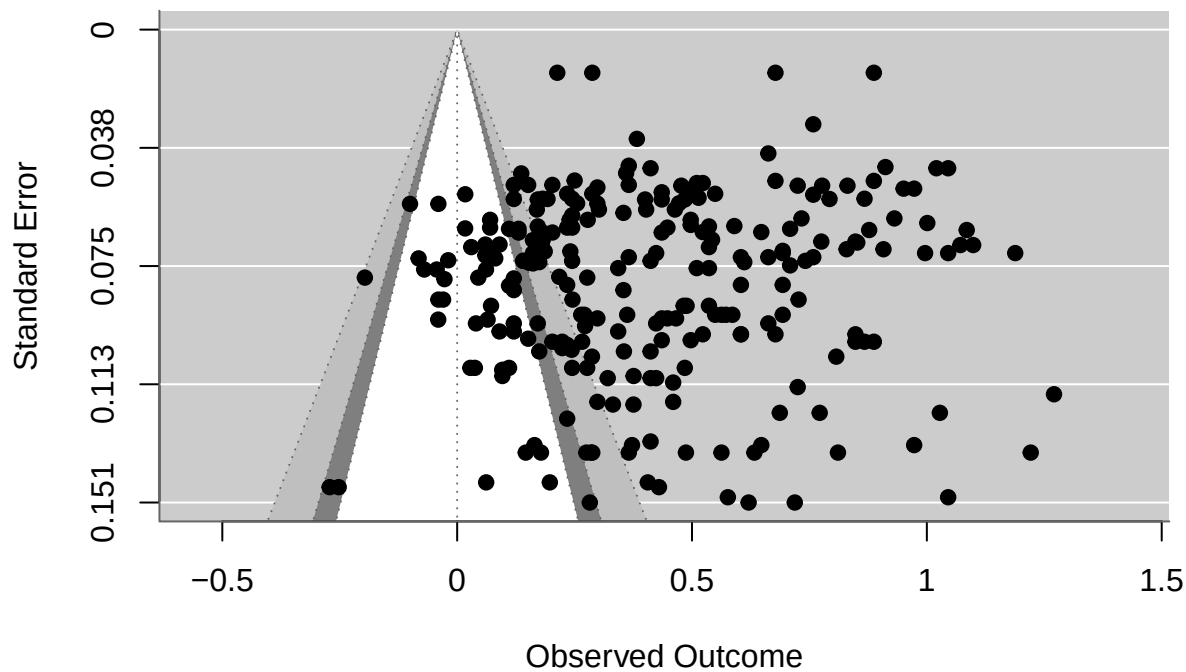
```
ex_unadj_mean_cil <- ex_unadj_mean$tau2 - 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_ciu <- ex_unadj_mean$tau2 + 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_cil; ex_unadj_mean_ciu
```

```
## [1] 0.06954973
```

```
## [1] 0.1020752
```

Forest plot

```
funnel(ex_unadj_mean, refline = 0, level = c(90, 95, 99))
```



CHE estimate: mean only

```
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
```

```r
    obs = esid,
    data = dat_ex,
    rho = 0.5,
    sparse = TRUE
  )

ex_che <- rma.mv(yi = yi,
                 V = V_mat,
                 random = ~ 1 | studyid / esid,
                 data = dat_ex,
                 sparse = TRUE,
                 digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)

ex_che
```

```
##
## Multivariate Meta-Analysis Model (k = 251; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.054  0.233    128     no       studyid
## sigma^2.2  0.034  0.185    251     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 250) = 7669.807, p-val < .001
##
## Number of estimates:   251
## Number of clusters:    128
## Estimates per cluster: 1-6 (mean: 1.96, median: 2)
##
## Model Results:
##
## estimate     se¹    tval¹      df¹   pval¹  ci.lb¹  ci.ub¹
##    0.404  0.025   16.005   124.32   <.001   0.354   0.454   ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

Adjusted estimate in metaselection: Beta model mean only

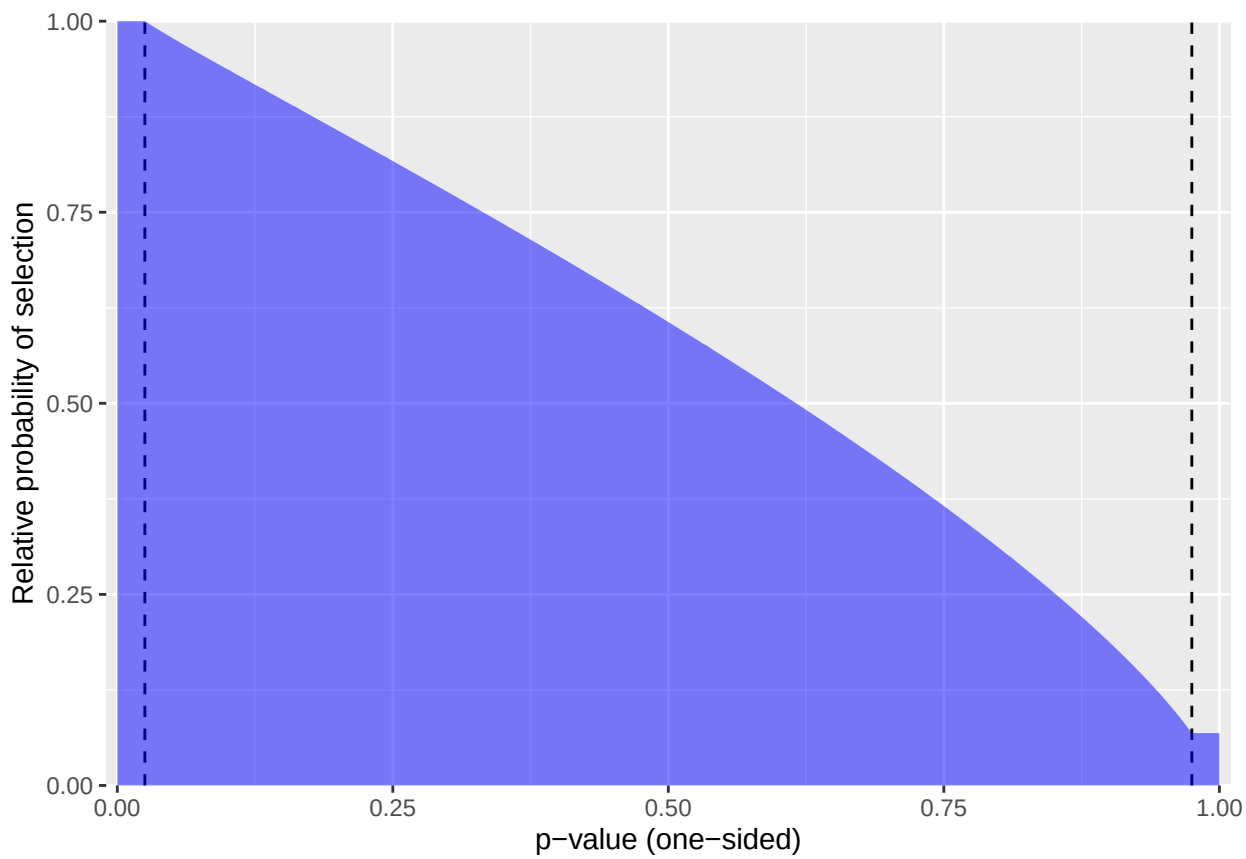
```r
ex_adj_ms_beta_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
```

```
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)
ex_adj_ms_beta_mean
```

```
##    param   Est     SE  p_value  CI_lo CI_hi
##     beta 0.279 0.0806 0.000534 0.1212 0.437
##     tau2 0.142 0.0255       NA 0.0997 0.202
##  lambda1 0.995 0.1392 0.971388 0.7564 1.309
##  lambda2 1.727 0.2490 0.000152 1.3016 2.291
```

Selection plot: Beta model mean only

```
selection_plot(ex_adj_ms_beta_mean)
```



Adjusted estimate in metaselection: Step 3PSM model mean only

```
ex_adj_ms_step3_mean <- selection_model(
  dat = dat_ex,
```

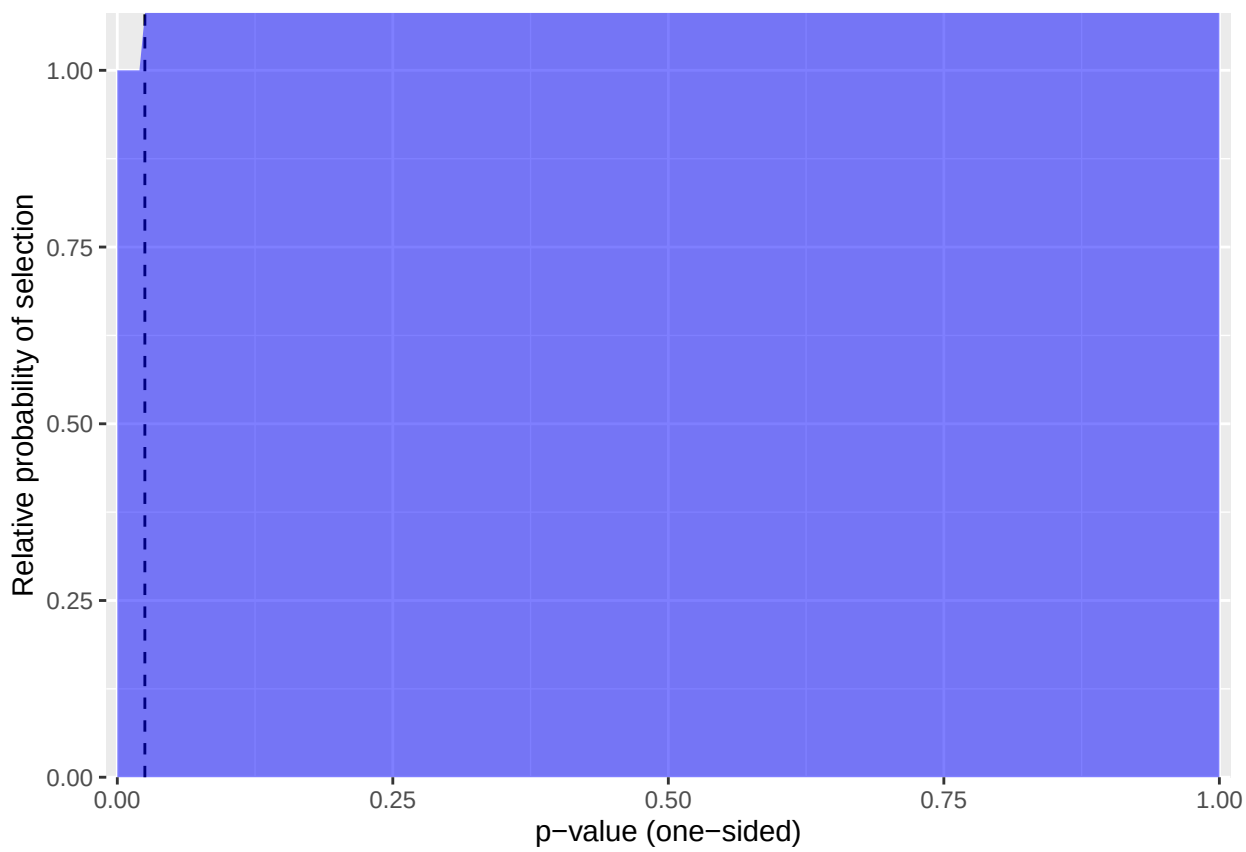
```
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

ex_adj_ms_step3_mean
```

```
##    param   Est      SE  p_value  CI_lo CI_hi
##     beta 0.421 0.03454 3.06e-34 0.3537 0.489
##     tau2 0.084 0.00982       NA 0.0668 0.106
##  lambda1 1.081 0.32392 7.94e-01 0.6010 1.945
```

Selection plot: Step 3 PSM model mean only

```
selection_plot(ex_adj_ms_step3_mean)
```



Adjusted estimate in metaselection: Step 4PSM model mean only

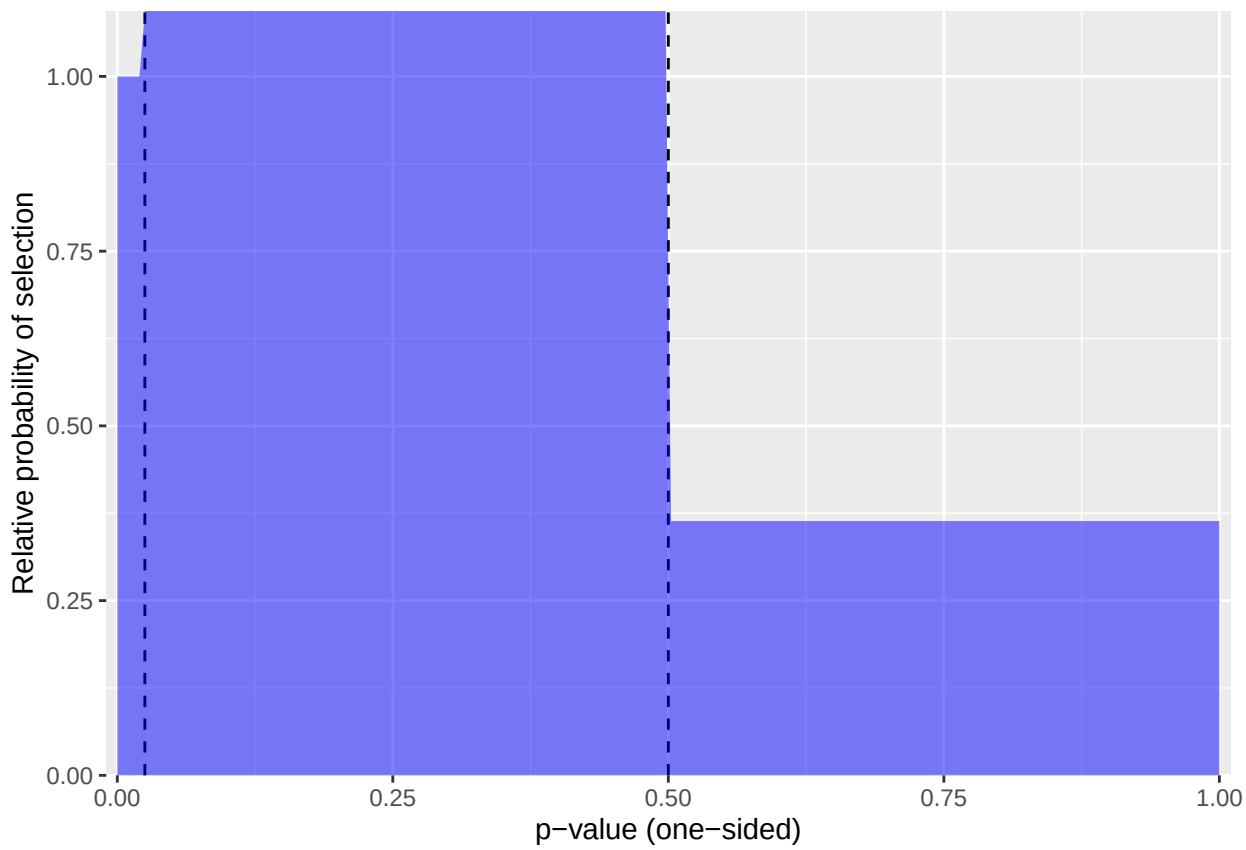
```r
ex_adj_ms_step4_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

ex_adj_ms_step4_mean
```

```
##    param   Est     SE  p_value  CI_lo CI_hi
##     beta 0.367 0.0462 2.07e-15 0.2761 0.457
##     tau2 0.107 0.0143       NA 0.0826 0.139
##  lambda1 1.094 0.2715 7.19e-01 0.6721 1.779
##  lambda2 0.364 0.1860 4.79e-02 0.1336 0.991
```

Selection plot: Step 4PSM model mean only

```r
selection_plot(ex_adj_ms_step4_mean)
```



Moderator

Unadjusted estimate: moderator

```
ex_unadj_mod <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML",
  mods = ~ design
)
ex_unadj_mod
```

```
##
## Mixed-Effects Model (k = 251; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.0743 (SE = 0.0073)
## tau (square root of estimated tau^2 value):             0.2726
## I^2 (residual heterogeneity / unaccounted variability): 95.35%
## H^2 (unaccounted variability / sampling variability):   21.52
## R^2 (amount of heterogeneity accounted for):            13.39%
##
## Test for Residual Heterogeneity:
## QE(df = 249) = 5627.9474, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 34.9749, p-val < .0001
##
## Model Results:
##
##          estimate      se    zval    pval   ci.lb   ci.ub
## intrcpt    0.2544  0.0326  7.7991  <.0001  0.1905  0.3183  ***
## design     0.2315  0.0391  5.9140  <.0001  0.1547  0.3082  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ex_unadj_mod_cil <- ex_unadj_mod$tau2 - 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_ciu <- ex_unadj_mod$tau2 + 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_cil; ex_unadj_mod_ciu
```

```
## [1] 0.06007707
```

```
## [1] 0.08856213
```

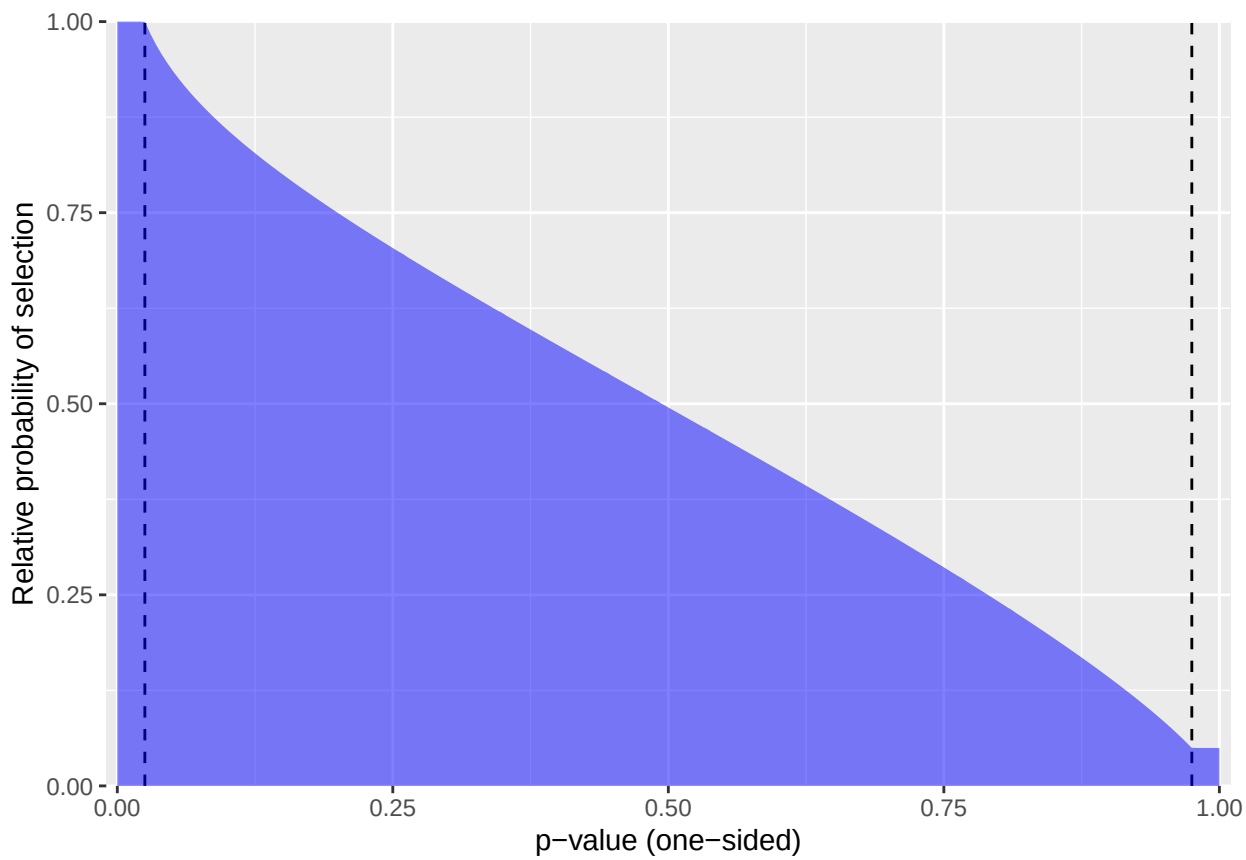
Adjusted estimate in metaselection: Beta model moderator

```r
ex_adj_ms_beta_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE,
  mean_mods = ~ design
)
ex_adj_ms_beta_mod
```

```
##             param     Est     SE  p_value   CI_lo CI_hi
##  beta_(Intercept) 0.00374 0.1192 0.974993 -0.2298 0.237
##       beta_design 0.39877 0.1062 0.000173  0.1907 0.607
##              tau2 0.11897 0.0191       NA  0.0868 0.163
##           lambda1 0.93320 0.1348 0.632080  0.7032 1.238
##           lambda2 1.75282 0.2701 0.000271  1.2958 2.371
```

Selection plot: Beta model moderator

```r
selection_plot(ex_adj_ms_beta_mod)
```



Adjusted estimate in metaselection: Step 3PSM model moderator

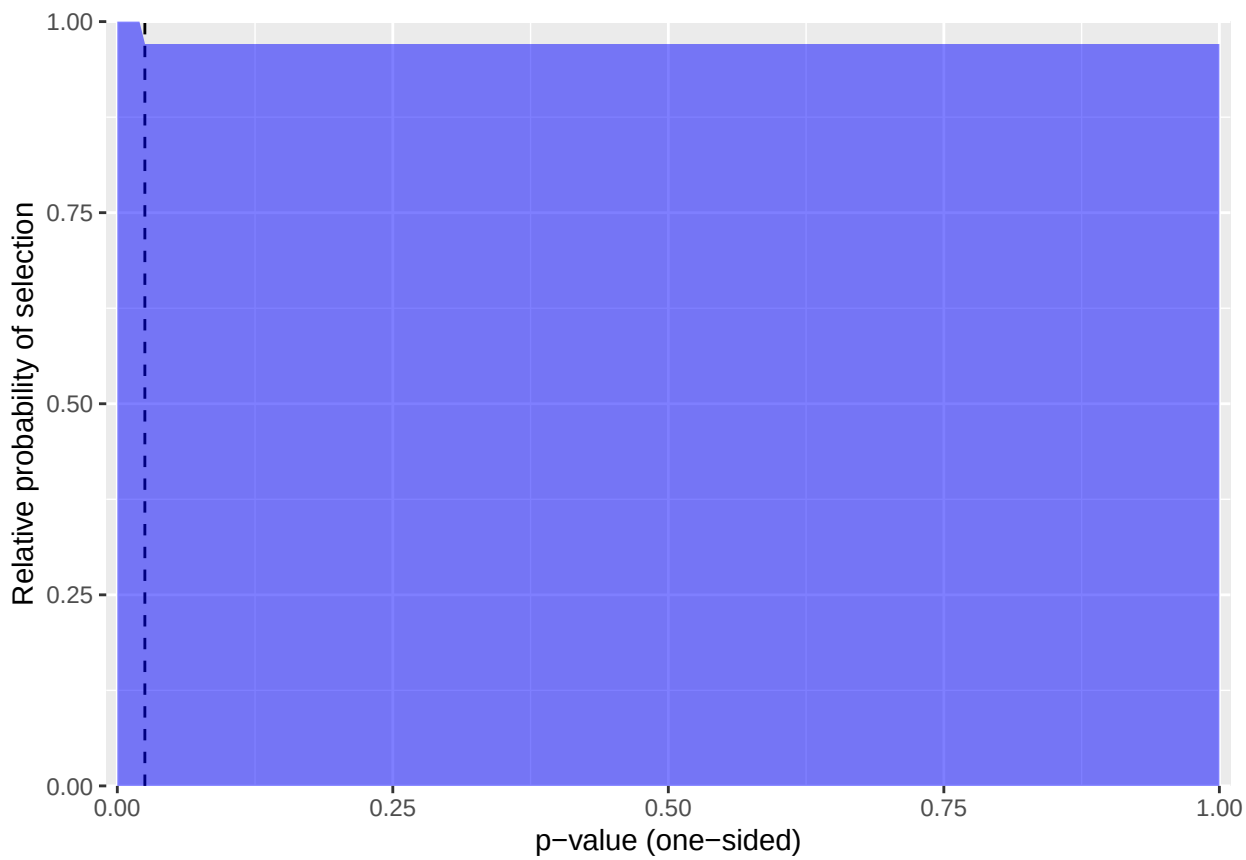
```
ex_adj_ms_step3_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025,
  mean_mods = ~ design
)

ex_adj_ms_step3_mod
```

```
##             param    Est      SE  p_value  CI_lo  CI_hi
##  beta_(Intercept) 0.2511 0.04510 2.59e-08 0.1627 0.3395
##       beta_design 0.2330 0.04944 2.44e-06 0.1361 0.3299
##              tau2 0.0746 0.00875       NA 0.0593 0.0939
##           lambda1 0.9707 0.29120 9.21e-01 0.5392 1.7476
```

Selection plot: Step 3 PSM model moderator

```
selection_plot(ex_adj_ms_step3_mod)
```



Adjusted estimate in metaselection: Step 4PSM model moderator

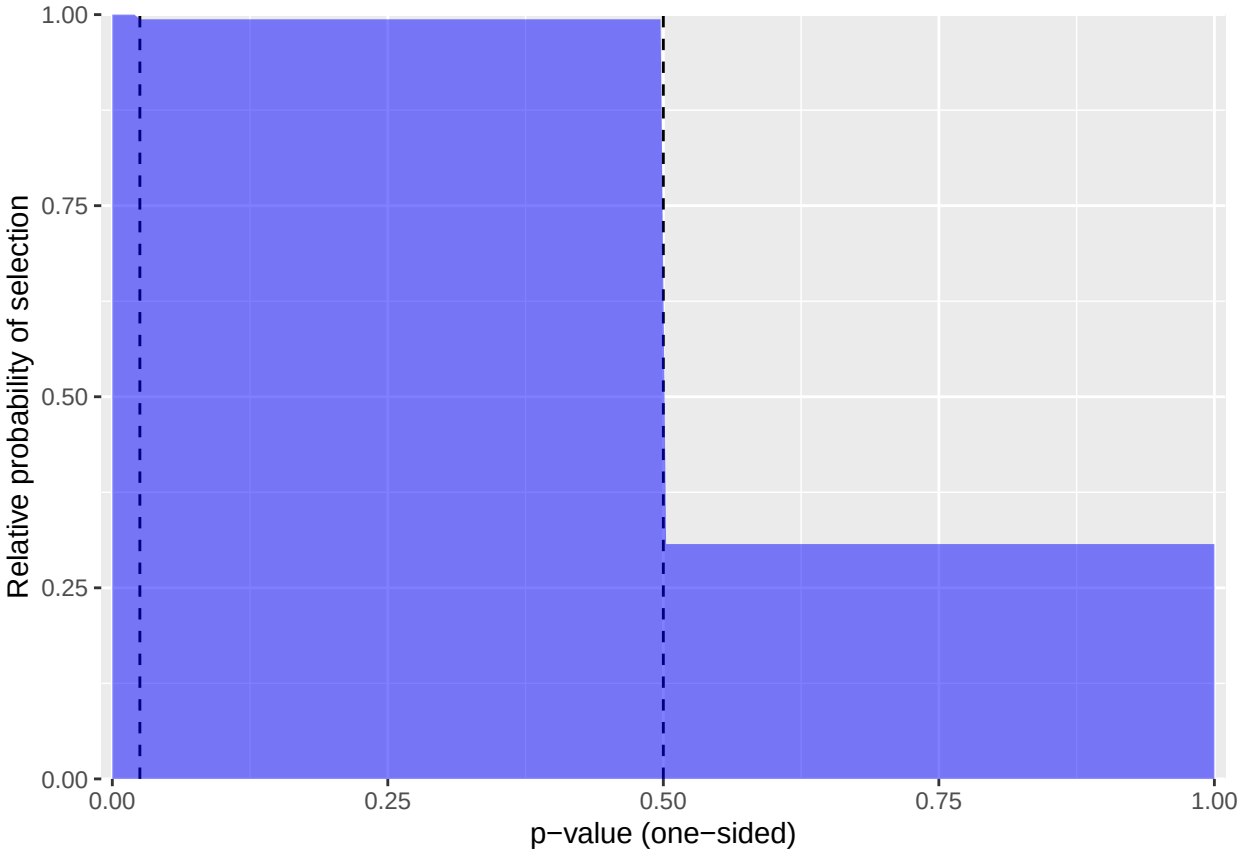
```
ex_adj_ms_step4_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5),
  mean_mods = ~ design
)

ex_adj_ms_step4_mod
```

```
##             param    Est     SE  p_value  CI_lo CI_hi
##  beta_(Intercept) 0.1545 0.0646 1.68e-02 0.0279 0.281
##       beta_design 0.2986 0.0660 6.12e-06 0.1692 0.428
##              tau2 0.0925 0.0123       NA 0.0713 0.120
##           lambda1 0.9939 0.2461 9.80e-01 0.6117 1.615
##           lambda2 0.3073 0.1616 2.48e-02 0.1096 0.861
```

Selection plot: Step 4PSM model moderator

```
selection_plot(ex_adj_ms_step4_mod)
```


1.00
0.75
0.50
0.25
0.00
Relative probability of selection
0.00
0.25
0.50
0.75
1.00
p−value (one−sided)